



Computing Infrastructures
July 10, 2024

Course Section:	☐ Prof. Ardagna	☐ Prof. Palermo	☐ Prof. Roveri
Student ID (Codice Persona):			
Last Name:	 (LAST NAME IN CAPITAL LETTERS)		
First Name:	 (FIRST NAME IN CAPITAL LETTERS)		

Exam Duration: 1hour and 30min

Students are not permitted to use mobile phones and similar connected devices. Course materials and programmable devices (e.g. programmable calculators) cannot be used as well. **Any violation of the rules is considered a cheating action.**

Answers must be given on the Answer Sheets and in English. Any box filled or answer provided on the other sheets will be ignored. Students must use a pen (black or blue) to mark the answers (no pencil).

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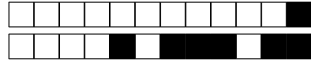
Check that the first number of the code for the Answer Sheet is the same as for the other sheets. The code can be found in the top-right corner of each page in the form +NN/KK/XX+. The parts that should correspond is **ONLY** the first digit NN.

Mark clearly the box corresponding to your answers, without overlapping on other boxes. If you make a mistake on them, circle the word *Question* together with the related number, and write the correct letter to its side.

Numerical exercises require writing the formulas and procedure used to solve the problem just after the question in the left space. Exercises without the procedure used to reach the result will not be considered for the evaluation. Only the numeric answer and its unit should be reported on the corresponding dotted line in the Answer Sheet.

The answers to the *Open Questions* should be written using ONLY the space available on in the boxes within the Answer Sheets. The answers should be readable by the professor. Unreadable answers will not be considered for the evaluation.

Scores: correct answers take positive points, unanswered questions take 0 points, **wrong answers can have negative points.** An indication of the points is available at the beginning of each section. The final score can be re-modulated at the end of the evaluation.



True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 Accelerators such as GPUs and TPUs are used in datacenters primarily to enhance general-purpose computing tasks.

☐ A False

☐ B True

Question 2 Leaf-spine topologies are commonly used in modern datacenters due to their efficiency and high-performance characteristics.

☐ A True

☐ B False

Question 3 FPGAs (Field-Programmable Gate Arrays) provide the flexibility to reconfigure hardware for specific tasks in datacenters.

☐ A False

☐ B True

Question 4 Virtualization helps to improve hardware utilization by enabling the efficient sharing of hardware resources.

☐ A True

☐ B False

Question 5 Warehouse-scale computers are known for their high fault tolerance and redundancy features.

☐ A False

☐ B True

Question 6 In-rack cooling involves placing cooling units between server racks.

☐ A True

☐ B False

Question 7 RAID 0 is best used for temporary data storage, such as caching, rather than long-term storage.

☐ A True

☐ B False

Question 8 Closed loop cooling systems are not suitable for high-density datacenters where heat dissipation is critical.

☐ A False

☐ B True

Question 9 Hypervisors cannot be used to run virtual machines on cloud computing platforms.

☐ A True

☐ B False

Question 10 RAID 5 can suffer from a performance penalty during a rebuild (data recovery in faulty disk).

☐ A False

☐ B True



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Consider an HDD with the following characteristics: block size 4KB, average seek time 3ms, data transfer time for a single block 0.5ms, and a negligible overhead controller. Knowing that the average locality is 75% and the average I/O service time to transfer an 800KB file is 400ms, what is the rotation speed for the disk in RPM?

Question 12

Consider a RAID 4 configuration composed of an array with 8 disks. What is the minimum number of I/O operations requested to update one block of a single data disk (considering the sum over the entire set of disks)?



Question 13

During the procurement of a server for an important scientific calculation, 3 different solutions have been offered.

- Server A allows to complete the target calculation in 400 hours, it has a MTTF of 1600 hours, and a MTTR of 3 hours;
- Server B allows to complete the target calculation in 500 hours, it has a MTTF of 1750 hours, and a MTTR of 4 hours.
- Server C allows to complete the target calculation in 600 hours, and it has a MTTR of 5 hours.

We know that the decision on which solution to buy depends on which server has the higher probability of completing the calculation before failure, once the calculation it is started. What should be the minimum MTTF for Server C to be selected as the system to buy? Use at least 4 decimal digit for each intermediate calculation.

Question 14

Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 8 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?



Question 15

A company wants to evaluate the performance of the services provided to its users. The computer system includes two servers $S1$ and $S2$. The system is considered as an open queue network model where the two servers work in tandem and the following measurements were obtained during 20-minute monitoring:

- Number of requests served at the system level: $C = 400$
- Number of requests served by server $S1$: $C_{S1} = 800$
- Number of requests served by server $S2$: $C_{S2} = 200$
- Busy time for server $S1$: $B_{S1} = 300$ sec
- Busy time for server $S2$: $B_{S2} = 900$ sec

What are the service demand and utilization for server $S1$ and server $S2$ (D_{S1} , D_{S2} , U_{S1} , U_{S2})?
 $D_{S1} = ?$ $D_{S2} = ?$ $U_{S1} = ?$ $U_{S2} = ?$

Question 16

Considering the same system as in the previous question 15, if you predict that your incoming workload is going to reach $\lambda = 3$ req/sec, what is the minimum number of instances for each type of server N_{S1} and N_{S2} that you need to introduce to keep their utilization less or equal to 60%? (Note1: when you introduce additional server instances at each layer of the tandem queue, you can assume that they equally split the number of visits across the server of the same type. Note 2: The service time of each server does not change while adding servers or increasing its workload.).
 $N_{S1} = ?$ $N_{S2} = ?$



+1/6/55+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

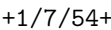
⇒ "SSD disks will replace HDDs in all the datacenters". Provide your opinion about this sentence and comment with details and examples.

Question 18

⇒ Rank and comment the most energy consuming aspects (both IT and not) in datacenters.

!!!ANY ANSWER PROVIDED ON THIS PAGE WILL BE IGNORED!!!

If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

[illegible]



●



Computing Infrastructures - July 10, 2024

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

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Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 : Disk Rotation Speed [RPM] =

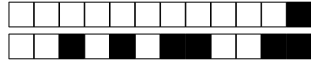
Question 12 : Number of I/O operations =

Question 13 : Minimum $MTTF_{ServerC}$ [hours] =

Question 14 : Maximum number of servers =

Question 15 : $Rmin_{10}$ = $Rmin_1$ =

Question 16 : N_{S1} = N_{S2} =



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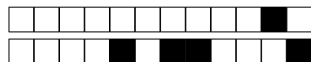
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**True false questions**

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☐ A True

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Question 4 FPGAs (Field-Programmable Gate Arrays) provide the flexibility to reconfigure hardware for specific tasks in datacenters.

☐ A False

☐ B True

Question 5 Warehouse-scale computers are known for their high fault tolerance and redundancy features.

☐ A True

☐ B False

Question 6 Closed loop cooling systems are not suitable for high-density datacenters where heat dissipation is critical.

☐ A True

☐ B False

Question 7 In-rack cooling involves placing cooling units between server racks.

☐ A True

☐ B False

Question 8 RAID 0 is best used for temporary data storage, such as caching, rather than long-term storage.

☐ A True

☐ B False

Question 9 Virtualization helps to improve hardware utilization by enabling the efficient sharing of hardware resources.

☐ A True

☐ B False

Question 10 Leaf-spine topologies are commonly used in modern datacenters due to their efficiency and high-performance characteristics.

☐ A True

☐ B False



+2/3/48+

Exercises

Correct answer: +2, No answer: 0.

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Question 11

Consider an HDD with the following characteristics: block size 4KB, average seek time 3ms, data transfer time for a single block 0.5ms, and a negligible overhead controller. Knowing that the average locality is 60% and the average I/O service time to transfer an 800KB file is 740ms, what is the rotation speed for the disk in RPM?

Question 12

Consider a RAID 4 configuration composed of an array with 8 disks. What is the minimum number of I/O operations requested to update one block of a single data disk (considering the sum over the entire set of disks)?



Question 13

During the procurement of a server for an important scientific calculation, 3 different solutions have been offered.

- Server A allows to complete the target calculation in 400 hours, it has a MTTF of 1200 hours, and a MTTR of 3 hours;
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Question 14

Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 6 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?



Question 15

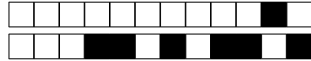
A company wants to evaluate the performance of the services provided to its users. The computer system includes two servers S_1 and S_2 . The system is considered as an open queue network model where the two servers work in tandem and the following measurements were obtained during 20-minute monitoring:

- Number of requests served at the system level: $C = 800$
- Number of requests served by S_1 : $C_{S1} = 800$
- Number of requests served by S_2 : $C_{S2} = 200$
- Busy time S_1 : $B_{S1} = 300$ sec
- Busy time S_2 : $B_{S2} = 450$ sec

What are the service demand and utilization for server $S1$ and server $S2$ (D_{S1} , D_{S2} , U_{S1} , U_{S2})?
 $D_{S1} = ?$ $D_{S2} = ?$ $U_{S1} = ?$ $U_{S2} = ?$

Question 16

Considering the same system as in the previous question 15, if you predict that your incoming workload is going to reach $\lambda = 3$ req/sec, what is the minimum number of instances for each type of server N_{S1} and N_{S1} that you need to introduce to keep their utilization less or equal to 70%? (Note1: when you introduce additional server instances at each layer of the tandem queue, you can assume that they equally split the number of visits across the server of the same type. Note 2: The service time of each server does not change while adding servers or increasing its workload.).
 $N_{S1} = ?$ $N_{S2} = ?$



+2/6/45+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

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Question 17

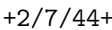
⇒ "SSD disks will replace HDDs in all the datacenters". Provide your opinion about this sentence and comment with details and examples.

Question 18

⇒ Rank and comment the most energy consuming aspects (both IT and not) in datacenters.

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If needed, you can use the space hereafter to organize your answer.



Answer Sheets (Page 1)

First Name (CAPITAL LETTERS):

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⇒ *"SSD disks will replace HDDs in all the datacenters"*. Provide your opinion about this sentence and comment with details and examples.



●



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Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 : Disk Rotation Speed [RPM] =

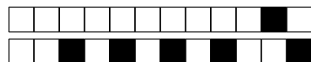
Question 12 : Number of I/O operations =

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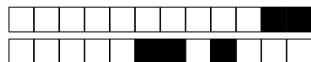
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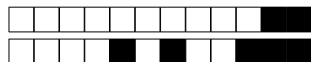
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True false questions

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☐ A False

☐ B True

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Question 9 Closed loop cooling systems are not suitable for high-density datacenters where heat dissipation is critical.

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Question 10 Accelerators such as GPUs and TPUs are used in datacenters primarily to enhance general-purpose computing tasks.

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Exercises

Correct answer: +2, No answer: 0.

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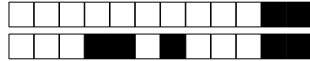
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 $N_{S1} = ?$ $N_{S2} = ?$



+3/6/35+

Open Questions

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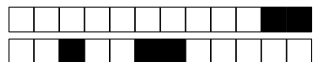
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+3/7/34+





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True/False Questions

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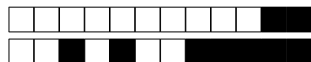
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July 10, 2024

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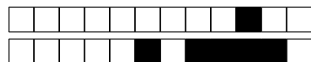
Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
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Computing Infrastructures
July 10, 2024

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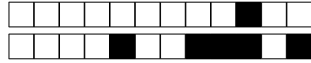
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True false questions

Correct answer: +1, No answer: 0, Wrong Answer -0.5

Answers must be given on the ANSWER SHEETS. Any box filled here will be ignored. Pay attention to the position (A or B) of the True/False answers, since they are not always in the same position.

Question 1 RAID 0 is best used for temporary data storage, such as caching, rather than long-term storage.

☐ A False

☐ B True

Question 2 Leaf-spine topologies are commonly used in modern datacenters due to their efficiency and high-performance characteristics.

☐ A True

☐ B False

Question 3 Warehouse-scale computers are known for their high fault tolerance and redundancy features.

☐ A False

☐ B True

Question 4 Closed loop cooling systems are not suitable for high-density datacenters where heat dissipation is critical.

☐ A False

☐ B True

Question 5 Accelerators such as GPUs and TPUs are used in datacenters primarily to enhance general-purpose computing tasks.

☐ A True

☐ B False

Question 6 RAID 5 can suffer from a performance penalty during a rebuild (data recovery in faulty disk).

☐ A True

☐ B False

Question 7 In-rack cooling involves placing cooling units between server racks.

☐ A False

☐ B True

Question 8 FPGAs (Field-Programmable Gate Arrays) provide the flexibility to reconfigure hardware for specific tasks in datacenters.

☐ A True

☐ B False

Question 9 Hypervisors cannot be used to run virtual machines on cloud computing platforms.

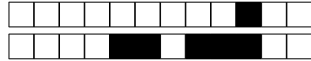
☐ A False

☐ B True

Question 10 Virtualization helps to improve hardware utilization by enabling the efficient sharing of hardware resources.

☐ A False

☐ B True



Exercises

Correct answer: +2, No answer: 0.

The formulas and procedures used to solve the exercises should be included here close to the question. The numeric answer, and only that, must be given on the ANSWER SHEETS. Any number written only here will be ignored. The correct number is ONLY a necessary condition for a correct answer. If the formulas are not available after each exercise, they will be considered as not answered.

Question 11

Consider an HDD with the following characteristics: block size 4KB, average seek time 3ms, data transfer time for a single block 0.5ms, and a negligible overhead controller. Knowing that the average locality is 60% and the average I/O service time to transfer an 800KB file is 740ms, what is the rotation speed for the disk in RPM?

Question 12

Consider a RAID 6 configuration composed of an array with 8 disks. What is the minimum number of I/O operations requested to update one block of a single data disk (considering the sum over the entire set of disks)?



Question 13

During the procurement of a server for an important scientific calculation, 3 different solutions have been offered.

- Server A allows to complete the target calculation in 400 hours, it has a MTTF of 1200 hours, and a MTTR of 3 hours;
- Server B allows to complete the target calculation in 500 hours, it has a MTTF of 1750 hours, and a MTTR of 4 hours.
- Server C allows to complete the target calculation in 600 hours, and it has a MTTR of 5 hours.

We know that the decision on which solution to buy depends on which server has the higher probability of completing the calculation before failure, once the calculation it is started. What should be the minimum MTTF for Server C to be selected as the system to buy? Use at least 4 decimal digit for each intermediate calculation.

Question 14

Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 8 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?



Question 15

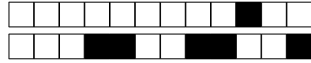
A company wants to evaluate the performance of the services provided to its users. The computer system includes two servers S_1 and S_2 . The system is considered as an open queue network model where the two servers work in tandem and the following measurements were obtained during 20-minute monitoring:

- Number of requests served at the system level: $C = 400$
- Number of requests served by S_1 : $C_{S1} = 800$
- Number of requests served by S_2 : $C_{S2} = 200$
- Busy time S_1 : $B_{S1} = 600$ sec
- Busy time S_2 : $B_{S2} = 850$ sec

What are the service demand and utilization for server $S1$ and server $S2$ (D_{S1} , D_{S2} , U_{S1} , U_{S2})?
 $D_{S1} = ?$ $D_{S2} = ?$ $U_{S1} = ?$ $U_{S2} = ?$

Question 16

Considering the same system as in the previous question 15, if you predict that your incoming workload is going to reach $\lambda = 3$ req/sec, what is the minimum number of instances for each type of server N_{S1} and N_{S1} that you need to introduce to keep their utilization less or equal to 60%? (Note1: when you introduce additional server instances at each layer of the tandem queue, you can assume that they equally split the number of visits across the server of the same type. Note 2: The service time of each server does not change while adding servers or increasing its workload.).
 $N_{S1} = ?$ $N_{S2} = ?$



+4/6/25+

Open Questions

Correct answer: +5, No answer: 0. Points are modulated considering the written text

Write the answer using ONLY the space available in the boxes on the ANSWER SHEETS. The answers should be readable by the professor. Unreadable answers will be considered wrong.

Question 17

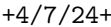
⇒ "SSD disks will replace HDDs in all the datacenters". Provide your opinion about this sentence and comment with details and examples.

Question 18

⇒ Rank and comment the most energy consuming aspects (both IT and not) in datacenters.

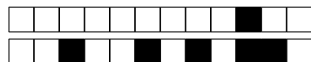
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If needed, you can use the space hereafter to organize your answer.





●



Computing Infrastructures - July 10, 2024

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 : Disk Rotation Speed [RPM] =

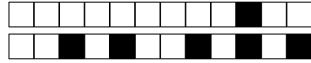
Question 12 : Number of I/O operations =

Question 13 : Minimum $MTTF_{ServerC}$ [hours] =

Question 14 : Maximum number of servers =

Question 15 : $Rmin_{10}$ = $Rmin_1$ =

Question 16 : N_{S1} = N_{S2} =



+4/10/21+

Computing Infrastructures
July 10, 2024

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

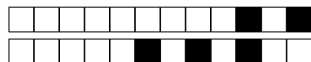
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+5/1/20+

Computing Infrastructures

July 10, 2024

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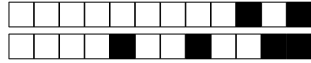
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☐ A True

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Question 3 Closed loop cooling systems are not suitable for high-density datacenters where heat dissipation is critical.

☐ A False

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Question 4 Leaf-spine topologies are commonly used in modern datacenters due to their efficiency and high-performance characteristics.

☐ A True

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Question 5 Hypervisors cannot be used to run virtual machines on cloud computing platforms.

☐ A False

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Question 6 RAID 0 is best used for temporary data storage, such as caching, rather than long-term storage.

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Question 7 Accelerators such as GPUs and TPUs are used in datacenters primarily to enhance general-purpose computing tasks.

☐ A False

☐ B True

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☐ A False

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Question 9 In-rack cooling involves placing cooling units between server racks.

☐ A True

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Exercises

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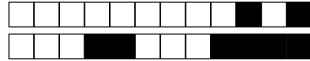
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Considering the same system as in the previous question 15, if you predict that your incoming workload is going to reach $\lambda = 3$ req/sec, what is the minimum number of instances for each type of server N_{S_1} and N_{S_2} that you need to introduce to keep their utilization less or equal to 60%? (Note1: when you introduce additional server instances at each layer of the tandem queue, you can assume that they equally split the number of visits across the server of the same type. Note 2: The service time of each server does not change while adding servers or increasing its workload.).
 $N_{S_1} = ?$ $N_{S_2} = ?$



+5/6/15+

Open Questions

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Question 17

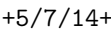
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●



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Answer Sheets (Page 3)

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True/False Questions

Question 01 : ☐A ☐B

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Exercises

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+6/1/10+

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☐ B True

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☐ B False

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☐ A False

☐ B True



+6/3/8+

Exercises

Correct answer: +2, No answer: 0.

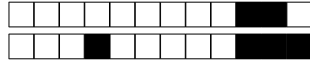
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Question 14

Using a two-tier leaf-spine topology without oversubscription and adopting only switches with 6 ports (all switches have the same number of ports), what is the maximum number of servers that can be connected?



Question 15

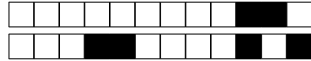
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- Number of requests served by server $S2$: $C_{S2} = 200$
- Busy time for server $S1$: $B_{S1} = 300$ sec
- Busy time for server $S2$: $B_{S2} = 900$ sec

What are the service demand and utilization for server $S1$ and server $S2$ (D_{S1} , D_{S2} , U_{S1} , U_{S2})?
 $D_{S1} = ?$ $D_{S2} = ?$ $U_{S1} = ?$ $U_{S2} = ?$

Question 16

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+6/6/5+

Open Questions

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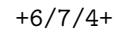
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Answer Sheets (Page 1)

Student ID (Codice Persona):

⇒ "SSD disks will replace HDDs in all the datacenters". Provide your opinion about this sentence and comment with details and examples.

[illegible]



●



Computing Infrastructures - July 10, 2024

Answer Sheets (Page 3)

Student ID (Codice Persona):

True/False Questions

Question 01 : ☐A ☐B

Question 02 : ☐A ☐B

Question 03 : ☐A ☐B

Question 04 : ☐A ☐B

Question 05 : ☐A ☐B

Question 06 : ☐A ☐B

Question 07 : ☐A ☐B

Question 08 : ☐A ☐B

Question 09 : ☐A ☐B

Question 10 : ☐A ☐B

Exercises

Question 11 : Disk Rotation Speed [RPM] =

Question 12 : Number of I/O operations =

Question 13 : Minimum $MTTF_{ServerC}$ [hours] =

Question 14 : Maximum number of servers =

Question 15 : $Rmin_{10}$ = $Rmin_1$ =

Question 16 : N_{S1} = N_{S2} =



+6/10/1+

Computing Infrastructures
July 10, 2024

Course Section: ☐ Prof. Ardagna ☐ Prof. Palermo ☐ Prof. Roveri

Student ID (Codice Persona):

Last Name:
(LAST NAME IN CAPITAL LETTERS)

First Name:
(FIRST NAME IN CAPITAL LETTERS)

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If needed, you can use this page for notes. Any answer written here will be ignored.